Movement Disorders

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Introduction

Movement disorders, such as Parkinson disease (PD), tremor, tics and dystonia, are **common** conditions.

The clinical presentation of movement disorders is **complex, often variable, and sometimes even bizarre.** Establishing the correct diagnosis can, therefore, be **difficult**.

Box 1 | Main categories of movement disorders

Insufficient movement

Akinetic, hypokinetic or bradykinetic syndromes

Too much movement (hyperkinesias or dyskinesias)

Jerky movements

- Myoclonus (including excessive startle)
- Chorea (including ballism)
- Tic disorders

Non-jerky movements

- Dystonia (including athetosis)
- Tremor

Akinetic–Rigid Syndromes

Akinesia is an **umbrella term** for a symptom complex that can include **bradykinesia** (slowness of movement) and **hypokinesia** (poverty of movement, and movements that are smaller than intended) and the **progressive fatiguing and decrement of repetitive alternating movements.**

In the Queen Square Brain Bank Criteria for the diagnosis of parkinsonism, bradykinesia is defined as including fatiguing and decrement of repetitive alternating movements.

Slowness of movement, without fatiguing and decrement, is seen in **pyramidal and cerebellar dysfunction** (often with additional clumsiness or irregularity).

Akinetic–Rigid Syndromes

An additional component of akinesia is absence or poverty of automatic movements, manifested by, for example, **hypomimia** with reduced blinking.

Increased muscle tone across a joint due to rigidity or spasticity can be differentiated while examining the full range of motion of a joint at varying speeds.

In **rigidity**, the **resistance is stable** and **equal** between flexion and extension movements, during the whole trajectory.

In **spasticity,** the tone is preferentially increased in **arm flexors and leg extensors,** and sudden decreases of muscle resistance (the '**clasp-knife phenomenon**') can be felt.

Jerky Hyperkinetic Syndromes

The Jerky hyperkinetic syndromes include **myoclonus (together with excessive startle), chorea and tics.**

Jerky movements might be seen in isolation or in combination with non-jerky movements.

Myoclonus

Myoclonus is a brief, rapid (<100 ms), **shock-like**, jerky movement consisting of single or repetitive muscle discharges.

Usually **positive** (caused by muscle contraction)

Can be negative (due to brief loss or inhibition of muscular tonus, as in asterixis—for example, when caused by hepatic encephalopathy [‘liver flap’]—or in uremic encephalopathy).

Negative myoclonus while walking produces a typical veering gait pattern, or the sudden postural lapses (‘bouncy gait’) seen in **postanoxic myoclonus.**

Myoclonus

Myoclonic muscle contractions are mostly accompanied by some **movement of the affected** body segment, in contrast to, for example, **fasciculations or myokymia,** where the twitches remain **within the affected** body segment.

When myoclonus **occurs in series,** the timing of the jerks can be either **rhythmic or irregular.**

Sometimes rhythmic myoclonus can be **mistaken for tremor** (examples include spinal segmental myoclonus, and hereditary cortical myoclonus, which has also been erroneously labeled ‘cortical tremor’).

Myoclonus

The distribution of myoclonus can be **focal, multifocal, segmental or generalized.**

Etiologically, myoclonus can be subdivided into **physiological myoclonus (for example, hypnic jerks), essential myoclonus (idiopathic or hereditary), epileptic myoclonus, or symptomatic myoclonus** in cases where the myoclonus is secondary to an underlying disorder.

Myoclonus

Physiologically, myoclonus is subdivided into **cortical, subcortical, spinal and peripheral types.**

Myoclonus can occur **spontaneously** (at rest), but is also often present—and **usually worsened—during movement (action myoclonus)**, or can be provoked by external tactile or acoustic stimuli (**reflex myoclonus**).

Myoclonic jerks differ from tics in that they are **not typically repetitive,** can severely interfere with normal voluntary movement, and **are not suppressible.**

Myoclonus

Cortical myoclonus is usually **action sensitive or stimulus sensitive,** mostly occurring in response to distal touch or stretch, and occasionally to visual stimuli.

Brainstem myoclonus, by contrast, is more commonly **provoked by auditory stimuli, or by tactile stimuli** around the face or snout.

Cortical myoclonus tends to be focal, whereas subcortical myoclonus is more often generalized.

The various types of myoclonus have differing neurophysiological characteristics.

Myoclonus

Propriospinal myoclonus; that is, myoclonus generated **within the spinal** cord with subsequent upward and downward spread. It's **difficult to diagnose,** as these movements are **slow and insufficiently jerky** in character. **Polymyographic** recordings are needed to prove the myoclonic nature of these axial movements.

Startle reactions, which are part of the spectrum of myoclonus, are also **provoked by external stimuli**—most often by auditory triggers, but also by surprise, alarm or acute pain. the startle reaction is characterized by a **bilaterally synchronous shock-like set of movements.**

Myoclonus

Reversible myoclonus can be seen with **metabolic disturbances** (renal failure, electrolyte imbalance, hypocalcemia), toxins, and many medications.

Essential myoclonus is a relatively **benign familial** condition characterized by **multifocal, very brief, lightning-like** movements that are frequently **alcohol-sensitive.**

Myoclonus

Treatment -- Managing the underlying condition or removing an offending agent.

Pharmacologic therapy --

GABAergic agents -- valproic acid (800–3000 mg/d), piracetam (8–20 g/d), clonazepam (2–15 mg/d), levetiracetam (1000–3000 mg/d), or primidone (500–1000 mg/d).

The **serotonin precursor 5-hydroxytryptophan (plus carbidopa)** may be useful in some cases of postanoxic myoclonus.



Video 1. Focal Cortical Myoclonus



Video 2. Myoclonus Epilepsy with demyelinating peripheral neuropathy and preserved intellect

Tics

A Tic is a **brief, rapid, recurrent, and seemingly purposeless stereotyped** motor contraction.

Difference from myoclonus and chorea is that patients report that their tics are **preceded by** rising discomfort or urge (‘sensory tic’) that is **relieved by** the actual movement (‘itch and scratch’ analogy).

Tics can be **suppressed** for short periods.

However, suppression of tics typically comes at the expense of mounting inner tension, leading to a ‘**rebound**’ of tics afterwards.

Tics

Tics usually **predominate** in the face, upper arms and neck.

They can be divided into **simple tics**, movement only affecting an individual muscle group, (for example, eye blinking, nose wrinkling, shoulder shrugging or throat clearing)

Complex tics, coordinated involvement of multiple muscle groups, (for example, touching things, smelling objects, echopraxia or jumping).

Another subdivision is into **motor tics** (such as stereotyped head jerks) or **phonic tics** (repetitive sniffs or sounds, words or even sentences).

Tics

Tics may be **less prominent or absent** during clinical examination.

A **motor tic** can have an **abrupt onset,** but the subsequent movement or posture might be slow or prolonged rather than jerky. This phenomenon is referred to as a **dystonic tic**, but the suppressibility and stereotyped nature are the clinical clues to classifying such movements as a tic.

Dystonic tics can occur in conjunction with other, non-dystonic tics.



Video 3. Sever Motor and Verbal Tics

Stereotypic Movements

Stereotypic movements resemble tics, but the actions consist of a complex set of movements that are **longer lasting, patterned, repetitive, purposeless and/or ritualistic.**

The movements can be **simple** (composed of a few maneuvers; for example, rocking or head banging) or **complex** (composed of multiple simple maneuvers performed together or in sequence).

Stereotypies are typically seen in patients with **autism, mental retardation, rett syndrome, psychosis, or congenital blindness and deafness.**



Video 4. Complex motor stereotypy (non-autistic)

TOURETTE'S SYNDROME (TS)

TS is a **neurobehavioral disorder.**

It predominantly affects **males**, and the **prevalence is 0.03–1.6%.**

TS is characterized by **multiple motor tics often accompanied by vocalizations (phonic tics).**

Patients characteristically can **voluntarily suppress tics for short periods of time**, but then experience an irresistible urge to express them.

Tics vary in intensity and may be absent for days or weeks only to recur, occasionally in a different pattern.

TOURETTE'S SYNDROME (TS)

Tics tend to present **between ages 2 and 15 years (mean 7 years)** and often disappear in adulthood, particularly in males.

Associated behavioral disturbances include **anxiety, depression, attention deficit hyperactivity disorder, and obsessive-compulsive disorder.**

Patients may experience **personality disorders, self-destructive behaviors, difficulties in school, and impaired interpersonal relationships.**

TOURETTE'S SYNDROME (TS)

Etiology and Pathophysiology

TS is thought to be a genetic disorder, but **no specific monogenic cause** has yet been identified.

The risk of a family with one affected child having a second is about 25%.

The pathophysiology of TS i**s not known**, but alterations in **dopamine neurotransmission, opioids, and second-messenger systems** have been proposed.



Video 5. TS with motor and phonic tics

Tics

TREATMENT

Mild disease often only require education and counseling.

Drug treatment is indicated when the tics are **disabling and interfere** with quality of life.

Therapy is **individualized.**

No singular treatment regimen.

α-agonist clonidine is used starting at low doses and gradually increasing the dose and frequency until satisfactory control is achieved.

Tics

Guanfacine (0.5–2 mg/d) is an α-agonist requires once-a-day dosing.

Neuroleptics may also be used.

Atypical neuroleptics are usually used (risperidone, olanzapine, ziprasidone) as there is **reduced risk of tardive dyskinesia**.

If they are not effective, low doses of **classical neuroleptics** such as haloperidol, fluphenazine, pimozide, or tiapride can be tried.

Tics

Tetrabenazine and deuterated tetrabenazine are being evaluated.

Botulinum toxin injections can be effective in controlling focal tics that involve small muscle groups.

Behavioral features, and particularly anxiety and compulsions should be treated appropriately.

DBS targeting the anterior portion of the internal capsule, the GPi, or the thalamus is currently under evaluation.

Chorea

Chorea is a greek word for **dance**.

It's a **jerky movement disorder**.

The 'choreography' includes a **constellation of randomly** flowing movements, which are, individually, jerky in nature.

Thus, chorea can be defined as involuntary movements that are **abrupt, unpredictable and nonrhythmic**, resulting from a continuous random flow of muscle contractions.

Other typical signs of chorea are **motor impersistence**, example fluctuating strength of the grip ('milkmaid's grip'), or **hung-up reflexes** (sustained contractions and choreatic movements after the knee-jerk reflex).

Chorea

Some patients with chorea—in HD, for example—can exhibit additional brief (<100 msec) **muscle jerks that are myoclonic**, and/or longer (>50msec) **co-contracting muscle spasms that are dystonic.**

Mild chorea may be **subtle and is difficult to detect.**

Choreatic patients have **few subjective complaints**, especially in early stages, usually their partner complains about the movements. (usually seen in HD)

Huntington's Disease

HD is a **progressive, fatal, highly penetrant autosomal dominant** disorder characterized by **motor, behavioral, oculomotor, and cognitive dysfunction.**

Onset --25 and 45 years (range, 3–70 years)

Prevalence -- 2–8 cases per 100,000

Average age at death -- 60 years.

It is **prevalent** in Europe, North America, South America, and Australia but is **rare** in African blacks and Asians.

Huntington’s Disease

Clinical Features

HD is characterized by **rapid, non patterned, semi-purposeful, involuntary choreiform movements**, and for this reason was formerly referred to as Huntington’s chorea.

However, **dysarthria, gait disturbance, oculomotor abnormalities, behavioral disturbance, and cognitive impairment with dementia** are also common features, thus the condition is currently referred to as HD.

Huntington's Disease

In the **early stages**, chorea tends to be **focal or segmental**, but progresses over time to involve multiple body regions.

With **advancing disease**, there tends to be a **reduction in chorea** and the **emergence of dystonia, rigidity, bradykinesia, and myoclonus.**

Depression with **suicidal tendencies, aggressive behavior, and psychosis** can be prominent features.

HD patients may also develop **noninsulin-dependent diabetes mellitus and neuroendocrine abnormalities (e.g., hypothalamic dysfunction).**

Huntington’s Disease

Diagnosis

A clinical diagnosis of HD can be **strongly suspected** in cases of chorea with a positive family history

Genetic testing is confirmatory.

The disease predominantly **affects the striatum** but progresses to involve the cerebral cortex and other brain regions.

Huntington’s Disease

Progressive atrophy of the head of the caudate nucleus, which form the lateral margin of the lateral ventricle, can be visualized by MRI, but the **putamen can be equally or even more severely affected.**

The neuropathology of HD consists of prominent **neuronal loss and gliosis** in the caudate nucleus and putamen; similar changes are also widespread in the cerebral cortex.

Intraneuronal inclusions containing aggregates of **ubiquitin** and the mutant protein **huntingtin** are found in the nuclei of affected neurons.

Huntington’s Disease

ETIOLOGY

HD is caused by an **increase** in the number of **polyglutamine (CAG) repeats (>40)** in the coding sequence of the **huntingtin gene** located on the **short arm of chromosome 4.**

The **larger the number** of repeats, the **earlier the disease** is manifest.

Acceleration of the process tends to occur with subsequent generations having larger numbers of repeats and earlier age of disease onset, is referred to as **anticipation**.

Mitochondrial dysfunction has been demonstrated in the striatum and skeletal muscle of symptomatic and presymptomatic individuals.



Video 6. Huntington’s Disease

Huntington’s Disease

TREATMENT

No disease-modifying therapy and symptomatic treatment is **limited.**

Dopamine-blocking agents may control the choreatic movements.

Tetrabenazine (a presynaptic dopamine depleting agent) has been approved for the treatment of chorea, but can **cause secondary parkinsonism.**

Huntington's Disease

Deuterated tetrabenazine (AustedoTM) interferes with the metabolism of tetrabenazine and **avoids a high Cmax.**

It has **fewer dose-related side effects** than tetrabenazine, and therefore can be administered in higher doses with potentially superior clinical benefits.

Neuroleptics are generally **not recommended** because of their potential to induce other troubling movement disorders.

These drugs **may be used** however in patients with **severe and disabling chorea**.

Huntington's Disease

Depression and anxiety should be treated with appropriate antidepressant and antianxiety drugs and monitored for mania and suicidal ideations.

Psychosis can be treated with **atypical antipsychotics** such as clozapine (50–600 mg/d), quetiapine (50–600 mg/d), and risperidone (2–8 mg/d).

Other investigative approaches include anti glutamate agents, dopamine stabilizers, caspase inhibitors, neurotrophic factors, anti-inflammatory agents, transplantation of fetal striatal cells or stem cells, and DBS of the globus pallidus pars interna (GPi), but **none has as yet been demonstrated to have a beneficial effect in HD.**

Huntington’s Disease-like Disorders

A group of rare inherited conditions that can **mimic HD,** designated HD-like (HDL) disorders, have also been identified.

HDL-1, 2, and **4 (Most common)** are **autosomal dominant** conditions that typically present in **adulthood.**

Huntington's Disease-like Disorders

Types	Characteristics
HDL-1	Due to expansion of an octapeptide repeat in PRNP, onset of personality change in the third or fourth decade, followed by chorea, rigidity, myoclonus, ataxia, and epilepsy.
HDL-2	Manifests in the third or fourth decade with a variety of movement disorders, Acanthocytosis can sometimes be seen in these patients. caused by an abnormally expanded CTG/CAG trinucleotide repeat expansion in the junctophilin-3 (JPH3) gene.
HDL-4	Caused by expansion of trinucleotide repeats in TBP, condition is identical to spinocerebellar ataxia (SCA) and most patients present primarily with ataxia rather than chorea.

Other Choreas

Chorea can be seen in a number of **additional disorders** related to genetic mutations or other disease states. Such as **Mutations in the ADCY5 (adenylate cyclase 5) gene, Chorea-acanthocytosis (neuroacanthocytosis), etc.**

Chorea may also occur in association with a variety of infections and degenerative disorders as well as vascular diseases and hypo and hyperglycemia.

Other Choreas

Sydenham's chorea (St. Vitus's dance) often develops in association with prior exposure to group A streptococcal infection. It is characterized by the acute onset of choreiform movements and behavioral disturbances. Dopamine-blocking agents, valproic acid, and carbamazepine may be used.

Chorea may recur in later life, particularly in association with **pregnancy (chorea gravidarum) or treatment with sex hormones.**

Other Choreas

Systemic lupus erythematosus is the **most common** systemic disorder that is associated with chorea.

Chorea can also be seen with **hyperthyroidism, autoimmune disorders including Sjögren's syndrome, infectious disorders including HIV disease, metabolic alterations, and polycythemia rubra vera.**

Chorea in association with many **medications** (especially anticonvulsants, cocaine, CNS stimulants, estrogens, lithium and chronic L-Dopa treatment).

Hemiballism

Ballism is considered under the rubric of chorea because it **shares the same** pathophysiology and treatment.

Ballism is a **violent form** of choreiform movement composed of **wild, flinging, large-amplitude movements** most frequently affecting **proximal limb muscles** on one side of the body (hemiballism).

Hemichorea -- if the amplitude of the movement is small.

Monoballism -- the movements involve only one limb.

Bilateral ballistic movements, rare, mostly due to metabolic abnormalities.

Hemiballismus

Paraballism -- If both upper or lower limbs.

The movements may be so severe as to cause **exhaustion, dehydration, local injury**, and, in extreme cases, **death**.
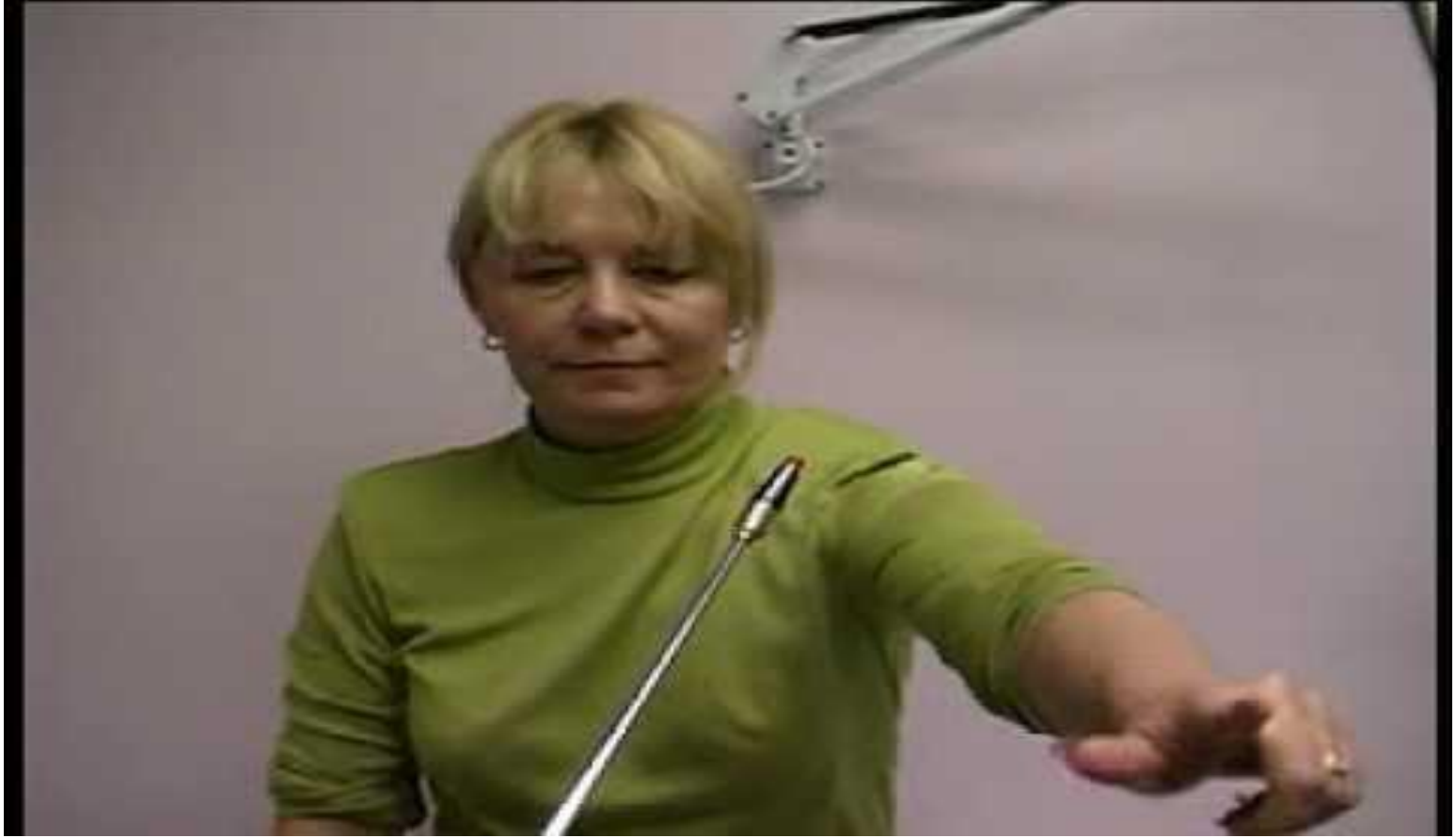
The most common cause is a partial lesion (infarct or hemorrhage) in the **subthalamic nucleus (STN),** but in 30–40% of cases the lesion is found in the **putamen, thalamus, or parietal cortex.**

Hemiballismus

Dopamine-blocking drugs can be very **helpful**.

Hemiballismus is usually **self-limiting** and resolves **spontaneously** after weeks or months.

In extreme cases, **pallidotomy or DBS of the GPi** can be effective and abolish the involuntary movements.



Video 7. Hemiballism

Non-jerky Hyperkinetic Syndromes

The non-jerky hyperkinetic syndromes include **tremor and dystonia.**

Tremors

Tremor consists of **alternating contractions** of agonist and antagonist muscles in an **oscillating, rhythmic manner.**

The movement does not necessarily involve a limb, as tremor can **affect any body part**, including the head, chin and soft palate.

The keyword in identifying tremor is **‘rhythmicity’**; that is, the oscillations occur at a **regular frequency.**

Tremors

Tremors have a **fixed frequency** and a **variable amplitude.**

Changes in amplitude -- may be spontaneous, may be result from movements or changes in posture, or from emotion and fatigue.

In patients with changes in tremor amplitude, use of **electromyography and accelerometry**, can confirm rhythmicity.

Table 1 | Classification of tremors according to moment of occurrence

Moment of occurrence	Features	Example of underlying disorder
A. At rest	Best judged in a body part that is fully supported against gravity	Parkinson disease
B. With action		
Postural	Occurs in body part that assumes a posture against gravity	Physiological; enhanced physiological (stress, endocrine disorders or intoxications); essential tremor
Kinetic		
Simple	Occurs during entire movement trajectory	Essential tremor
Intention	Progressively increases towards intended target	Cerebellar ataxia
Task specific	Occurs only during specific activities	Dystonic writing tremor
Isometric	Occurs during voluntary muscle contractions against a stationary resistance	Physiological; associated with other types of tremor
C. Combinations	Various	Severe essential tremor; atypical parkinsonism; dystonic tremor; rubral (Holmes) tremor

The above classification was proposed by a Consensus Statement of the Movement Disorder Society.[17]

Tremors

Mixed Tremors

Patients with PD may have **resting tremor** and an **associated postural tremor** with a higher frequency.

Holmes tremor (also known as **midbrain or rubral tremor**), which typically has **resting, postural and intention** components, **low frequency** of around 2–3Hz.

Tremor involving a body part affected by dystonia is classified, according to the Movement Disorder Society consensus statement, as "**Dystonic Tremor**'. For example, many patients with spasmodic torticollis also have a postural tremor of one or both arms.

Tremors

Dystonic tremor can mimic the tremor of PD.

People with dystonic tremor **do not** have true akinesia.

They also show **normal** dopamine transporter imaging.

‘Cogwheeling’ is jerky, brief contractions throughout the entire range of passive movement. Cogwheeling can be felt in patients with **essential or dystonic tremor, and PD.**

In the first two examples, however, there is **no additional rigidity** (in which case one speaks of ‘cogwheel rigidity’) or **akinesia**, whereas these features are present in PD.

Tremors

Physiologic tremor -- normal individuals, **distractible,** mild, high-frequency (10–12 Hz), postural or action tremor, affects the upper extremities.

Usually of no clinical consequence.

Enhanced physiologic tremor (EPT)-- affects 10% of the population, and occurs in association with anxiety, fatigue, a metabolic disturbance (hyperthyroidism, electrolyte abnormalities), drugs (valproate, lithium), or toxins (caffeine, smoking, alcohol).

Treatment -- control of any underlying disorder and beta blockers, if needed.

Essential Tremor

ET is the **most common** movement disorder.

High prevalence in those **aged >70 years.**

ET is characterized by a **high frequency tremor (6–10 Hz)** that predominantly affects the **upper extremities.**

May be a postural or action tremor.

Typically **bilateral and symmetric** but may begin on one side and remain asymmetric.

Essential Tremor

Severe ET can have an **intention tremor with overshoot and slowness of movement** suggesting the possibility of a **cerebellar origin.**

Tremor involves the **head in ~30%** of cases, voice in ~20%, tongue in ~20%, face/jaw in ~10%, and lower limbs in ~10%.

Multiple body parts are involved in at least 50% of cases.

The tremor is characteristically **improved by alcohol and worsened by stress.**

Essential Tremor

Subtle **impairment of coordination** or tandem walking may be present

Disturbances of hearing, cognition, personality, mood, and olfaction may be present

Usually the **neurologic examination is normal** aside from tremor.

The differential diagnosis includes **dystonic tremor or PD.**

Essential Tremor

- **PD**

1. Tremor stops at the onset of a voluntary action.
2. Typically associated with bradykinesia with progressive slowing of sequential movements (sequence effect)
3. Rigidity
4. Gait and postural instability, and other parkinsonian features.
5. Micrographia of PD.

- **ET**

1. Tremor rarely present at rest. (Emergent tremor -- rest tremor in ET that begin after a latency of a few seconds)
2. No associated symptoms of PD
3. Patients have relatively large handwriting with evidence of the effect of tremor.

Essential Tremor

ETIOLOGY AND PATHOPHYSIOLOGY

The etiology and pathophysiology of ET are **not known**.

50% of cases have a **positive family history with an AD pattern of inheritance.**

The **cerebellum and inferior olives** are possible sites of a "**tremor pacemaker**" based on the presence of cerebellar signs in about 10% of ET patients, and increased metabolic activity and blood flow in these regions in some patients.

Essential Tremor

TREATMENT

Mild cases - require **no treatment** other than reassurance.

Severe tremor (interference with eating, writing, etc) - **Beta blockers and primidone** are the standard drug therapies for ET.

Propranolol (20–120 mg daily, given in divided doses).

Primidone started at 12.5 mg and gradually increased (125–250 mg tid) to avoid sedation, nausea, and dizziness.

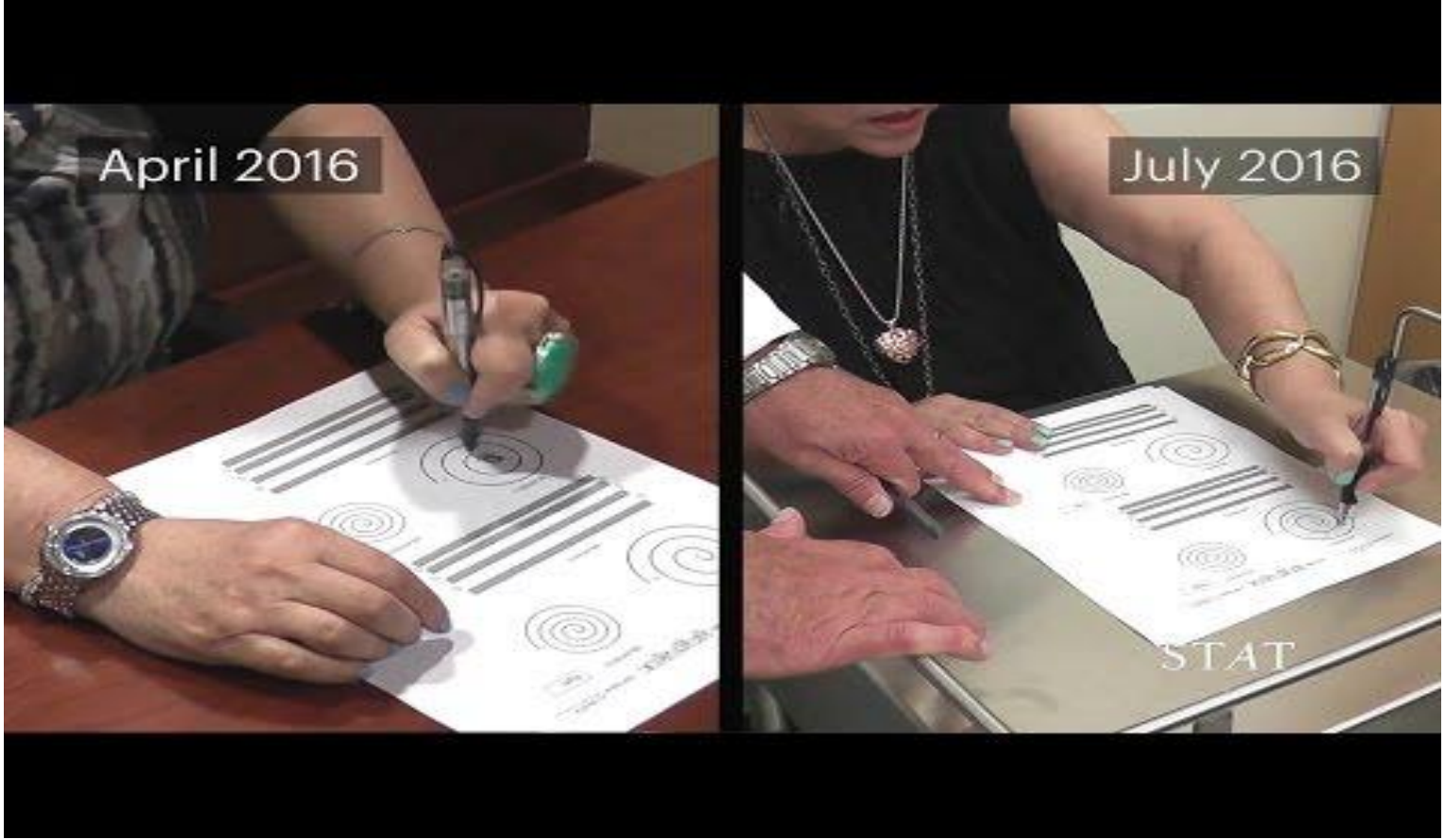
Essential Tremor

Gabapentin & Topiramate -- may be beneficial.

Botulinum toxin injections -- limb or voice tremor, but treatment can be associated with muscle weakness.

Severe and drug-resistant cases -- Surgical therapies targeting the **ventro-intermediate (VIM) nucleus of the thalamus** may be helpful.

Focal ultrasound -- Non Surgical alternative, under evaluation.



Video 8. Essential Tremor

Dystonia

Dystonia is a movement disorder characterized by sustained or intermittent muscle contractions of antagonist muscles causing **abnormal often repetitive movements and postures**.

Dystonic movements are typically **patterned and twisting**.

May be associated with a "**dystonic tremor**".

Initiated or worsened by voluntary action (action dystonia) and associated with **overflow** muscle activation.

Dystonia

Dystonia maybe **focal** minor contractions affecting only an **individual muscle** group.

Maybe severe and disabling involvement of **multiple muscle** groups.

Aggravated by **stress and fatigue.**

Attenuated by **relaxation and sensory tricks** such as touching the affected body part (geste antagoniste).

Dystonia

A Movement Disorder Society Task Force charged with redefining dystonia recommends classifying dystonia along two main axes: clinical and etiologic.

- Clinical -- dystonia can be categorized by
  - **Age of onset** (infancy, childhood, adolescence, early and late adulthood)
  - **Body distribution** (focal, segmental, multifocal, and generalized)
  - **Temporal pattern** (static or progressive, action-specific [diurnal and paroxysmal])
  - **Association with additional features.**

Clinical description along these lines enables formulating specific dystonia syndromes (e.g., early-onset generalized isolated dystonia).

Dystonia

Genetic features used for classification include **mode of inheritance or identification of a specific genetic defect.**

- **Isolated Dystonia --** Dystonia is the only disease manifestation with the exception of tremor.
- **Combined Dystonia --** Dystonia co-occurs with another movement disorder such as parkinsonism or myoclonus.
- **Complex dystonia --** Disorders in which dystonia is only one of several clinical manifestations and may be a less prominent or even inconsistent feature.

Isolated Dystonias

Focal (Multifocal, Segmental) Dystonia

Adult-onset, focal dystonia is the **most frequent** form of isolated dystonia

Women:men = 2:1.

Focal dystonia typically presents in the fourth to sixth decade and can be focal, multifocal, or segmental.

Isolated Dystonias

The **major** clinical phenotypes are as follows:

- **Cervical dystonia (50%)**—dystonic contractions of neck muscles causing the head to deviate to one side (laterocollis), twist (torticollis), move in a forward direction (anterocollis), or move in a backward direction (retrocollis). Muscle contractions can be painful and occasionally can be complicated with a secondary cervical radiculopathy.
- **Blepharospasm (20%)**—dystonic contractions of the eyelids with increased blinking that can interfere with reading, watching television, working on a computer, and driving. This can sometimes be so severe as to cause functional blindness.



Video 9. Severe cervical dystonia

Isolated Dystonias

- **Oromandibular dystonia (OMD) (1%)**—contractions of muscles of the lower face, lips, tongue, and jaw (opening or closing). **Meige's syndrome** is a combination of OMD and blepharospasm that predominantly affects women aged >60 years.
- **Spasmodic dysphonia (2%)**—dystonic contractions of the vocal cords during phonation, causing impaired speech. Most cases affect the adductor muscles and cause speech to have a choking or strained quality. Less commonly, the abductors are affected, leading to speech with a breathy or whispering quality.
- **Limb dystonias (5-7%)**—these can be present in either arms or legs and are often brought out by task-specific activities such as handwriting (writer's cramp), playing a musical instrument (musician's cramp), or putting in golf (the yips).



Video 10. Blepharospasm with Facial and Cervical Dystonia

Isolated Dystonias

Focal dystonias are often associated with a **high frequency tremor** that can resemble ET.

It can usually be distinguished from ET because it tends to occur in **conjunction with the dystonic contraction** and disappears when the dystonia is relieved (e.g., turning the head in the opposite direction of the dystonia).

Generalized Dystonia

Generalized dystonia is often **hereditary** in nature.

Age of onset in **childhood or adolescence.**

Genes commonly associated **TOR1A (Most Common), THAP1, ANO3, and GNAL.**

According to the recommendations of the International Parkinson's Disease and Movement Disorder Society, confirmed monogenic forms are classified according to absence or presence of accompanying clinical features and preceded by a "DYT" prefix, e.g., DYT-TOR1A.

Generalized Dystonia

Inheritance -- **Autosomal dominant.**

Found in **<5%** of dystonia patients.

Not all mutation carriers develop generalized dystonia.

About **35% remain unaffected.**

Rarely, they present with dystonia that **remains** focal or segmental in nature.

Combined Dystonia

Dystonia occurs **in conjunction** with a different movement disorder, such as parkinsonism or myoclonus.

Dopa-responsive dystonia (DRD; also known as **Segawa syndrome**)

Caused by mutations in the **GCH1 gene** (GTP cyclohydrolase-1) that **encodes for the rate-limiting enzyme** in the biosynthesis of dopamine via the **biopterin pathway.**

Childhood-onset form of dystonia with **diurnal fluctuations.**

Condition **dramatically responds** to low doses of **levodopa**.

Combined Dystonia

Parkinsonism can be a **major**, or even the only finding.

Presynaptic **dopaminergic deficit** as evidenced by SPECT.

Mutation penetrance of around 50% which is **higher in women** compared to men.

Recessively inherited (biallelic) mutations in GCH1 result in a much more **severe** clinical phenotype with developmental delay and infantile onset.

There is a **lifelong and dramatic response** to levodopa therapy.

All young onset forms of dystonia should be tested with **levodopa** to exclude the possibility of DRD.

Combined Dystonia

X-linked dystonia-parkinsonism (Lubag) is a combined form of dystonia and parkinsonism.

Exclusively in patients of **Filipino origin** due to a founder effect.

Patients develop **focal (cranial) dystonia** first that **rapidly generalizes** and, after 5–10 years, is gradually replaced by a form of **L-dopa unresponsive** parkinsonism.

The exact mutation causing X-linked dystonia-parkinsonism (Lubag) is **not yet known.**

TABLE 428-2 Confirmed Monogenic Forms of Isolated and Combined Dystonia[a]

FORM OF DYSTONIA		GENE	LOCUS NAME	DESIGNATION AND PHENOTYPIC SUBGROUP[a]	ADDITIONAL DISTINGUISHING FEATURES	MOI
Isolated		*TOR1A*	DYT1	DYT-TOR1A	Childhood or adolescent-onset, generalized	AD
		THAP1	DYT6	DYT-THAP1	Adolescent-onset, cranial or generalized	AD
		ANO3	DYT24	DYT-ANO3	Adult-onset, focal or segmental	AD
		GNAL	DYT25	DYT-GNAL	Mostly adult-onset, focal or segmental	AD
		KMT2B[b]	DYT28	DYT-KMT2B	Early-onset, generalized, mild syndromic features	AD
Combined	**Dystonia plus Parkinsonism**	*GCH1*	DYT5a	DYT-GCH1	Dopa-responsive	AD
		TAF1	DYT3	DYT-TAF1	Neurodegeneration	XL
		PRKRA	DYT16	DYT-PRKRA	Dystonia with mild parkinsonism	AR
		ATP1A3	DYT12	DYT-ATP1A3	Rapid-onset	AD
	Dystonia plus Myoclonus	*SGCE*	DYT11	DYT-SGCE	Psychiatric disease	AD

[a]According to C Marras et al: Mov Disord 31:436, 2016. [b]Several, but not all, patients show syndromic features; DYT-KMT2B may thus be better placed with the complex dystonias.

Complex Dystonias

Complex dystonias -- dystonia is **one part of a syndrome** with multiple different disease manifestations.

Maybe **hereditary** such as Wilson's disease (WD), Huntington's disease (HD), Lesch-Nyhan syndrome, corticobasal ganglionic disorders, and a variety of other neurologic, neurometabolic, and mitochondrial disorders.

Complex dystonias may also develop as a **consequence of drugs or toxins** (previously referred to as secondary dystonias).

Complex Dystonias

Drug-induced dystonia may be acute or chronic, and is most commonly seen with neuroleptic drugs or after chronic levodopa treatment in PD patients.

Dystonia can also be observed following **discrete lesions** in the striatum, and occasionally in the pallidum, thalamus, cortex, and brainstem due to infarction, hemorrhage anoxia, trauma, tumor, infection, or toxins such as manganese or carbon monoxide.

Dystonia can develop **following** peripheral nerve injury and be **associated with** features of complex regional pain syndrome.

A **psychogenic origin** is responsible for some cases of dystonia.

Pathophysiology Of Dystonia

The pathophysiologic basis of dystonia is **not completely known.**

The phenomenon is characterized by co-contracting synchronous bursts of agonist and antagonist muscle groups with recruitment of muscle groups that are not required for a given movement (**overflow**).

Physiologically, **loss of surround inhibition** is observed at multiple levels of the motor system (e.g., cortex, brainstem, spinal cord) accompanied by **increased cortical excitability and reorganization.**

Pathophysiology Of Dystonia

The **basal ganglia** is the site of origin of at least some types of dystonia because there are **alterations in blood flow and metabolism** in these structures.

Lesions of the basal ganglia **(particularly the putamen)** can induce dystonia, and surgical ablation or deep brain stimulation (DBS) of specific regions of the globus pallidus may ameliorate dystonia.

The **dopamine system** has also been implicated, because dopaminergic therapies can both induce and treat some forms of dystonia in different circumstances.

No specific pathology has been consistently identified in dystonia.

Treatment

Mostly symptomatic except in rare cases where correction of a primary underlying condition is possible.

Wilson's disease should be ruled out.

Levodopa should be tried in all cases of childhood- onset dystonia to test for DRD.

High-dose anticholinergics (trihexyphenidyl 20–120 mg/d) may be beneficial in **children**, but adults can rarely tolerate high doses because of **side effects** related to cognitive impairment and hallucinations.

Treatment

Oral baclofen (20–120 mg) benefits are usually **modest**, and **side effects** of sedation, weakness, and memory loss can be problematic.

Intrathecal infusion of baclofen is more likely to be useful, particularly for leg and trunk dystonia, but **benefits are frequently not sustained**, and **complications** can be serious and include infection, seizures, and coma.

Tetrabenazine (starting dose is 12.5 mg/d and treating dose is 25–75 mg/d), but its use may be limited by sedation and the development of parkinsonism.

Treatment

Neuroleptics are not recommended because of their potential to induce parkinsonism and other movement disorders, including tardive dystonia.

Clonazepam and diazepam are rarely effective.

Botulinum toxin is the preferred treatment for patients with focal dystonia, particularly in blepharospasm, torticollis, and spasmodic dysphonia.

Botulinum toxin blocks the release of acetylcholine at the neuromuscular junction.

Side effects are -- **excessive weakness,** particularly if it involves neck and swallowing muscles.

Treatment

Surgical therapy is for patients with severe dystonia who are **not responsive** to other treatments.

Peripheral procedures such as **rhizotomy and myotomy**, rarely employed.

DBS of the pallidum can provide **dramatic benefits**.

Benefits tend to be obtained with a **lower frequency** of stimulation and often occur after a relatively **longer latency** (weeks to months) than in PD.

Better results are typically obtained in **younger patients with shorter disease duration.**

Treatment

Supportive treatments such as **physical therapy and education** should be a part of the treatment regimen.

Dystonic Storm

Rare but potentially fatal condition that can occur in response to a **stress** situation such as surgery or a systemic infection in patients with preexisting dystonia.

There is **Acute onset of generalized and persistent dystonic contractions** that can involve the vocal cords or laryngeal muscles, leading to airway obstruction. There may be **rhabdomyolysis with renal failure**.

Treatment -- A combination of anticholinergics, diphenhydramine, baclofen, benzodiazepines, and dopaminergic agents. **Anesthesia with muscle paralysis** may be required.

Drug-induced Movement Disorders

Primarily associated with drugs that **block dopamine receptors (neuroleptics) or central dopaminergic transmission.**

Used in **psychiatry**, in the treatment of nausea or vomiting (e.g., prochlorperazine [Stemetil]) or gastroesophageal disorders (e.g., metoclopramide).

Hyperkinetic movement disorders secondary to neuroleptic drugs can be divided into those that present **acutely, subacutely, or after prolonged exposure (tardive syndromes).**

Drug-induced Movement Disorders

Dopamine-blocking drugs can be associated with a **reversible parkinsonian syndrome.**

Anticholinergics are often concomitantly prescribed, but this may increase the risk of developing a tardive syndrome.

Acute

Dystonia is the **most common acute** hyperkinetic drug reaction.

It is typically **generalized in children and focal in adults** (e.g., blepharospasm, torticollis, or OMD).

The reaction can develop **within minutes** of exposure.

The abrupt onset of severe spasms **may be confused with a seizure.**

The **acute onset** of chorea, stereotypic behavior, and tics may also be seen, particularly following exposure to **CNS stimulants** such as methylphenidate, cocaine, or amphetamines.

Acute

Treated with parenteral administration of anticholinergics (benztropine or diphenhydramine), benzodiazepines (lorazepam, clonazepam, or diazepam), or dopamine agonists.

Subacute

Akathisia is the most common reaction in this category.

It consists of **motor restlessness with a need to move** that is alleviated by movement.

Therapy consists of **removing** the offending agent.

When this is not possible, symptoms may be ameliorated with **benzodiazepines, anticholinergics, beta blockers, or dopamine agonists.**

Tardive Syndromes

These disorders develop **months to years** after initiation of the neuroleptic agent.

Tardive dyskinesias (TD) are **most common,** and typically present with choreiform and/or dystonic movements involving the mouth, lips, and tongue. In severe cases, the trunk, limbs, and respiratory muscles may also be affected.

TD usually **remits within 3 months** of stopping the drug, and most patients gradually improve over the course of several years.

The movements are often mild but they can be **severe and disabling**, particularly in the context of an **underlying psychiatric disorder.**

Tardive Syndromes

Atypical antipsychotics (e.g., clozapine, risperidone, olanzapine, quetiapine, ziprasidone, and aripiprazole) are associated with a lower risk of causing TD in comparison to traditional antipsychotics.

Younger patients have a **lower risk.**

Elderly, females, and those with underlying organic cerebral dysfunction have a **greater risk.**

Chronic use is associated with increased risk.

Tardive Syndromes

TD can be **permanent and resistant** to treatment.

Treatment **primarily consists of stopping** the offending agent.

Abrupt cessation can induce worsening.

TD can persist after withdrawal of antipsychotics and can be difficult to treat.

Tardive Syndromes

Valbenazine (IngrezzaTM) is an ester of tetrabenazine that has been approved for the treatment of tardive dyskinesia, but it is associated with **sleepiness and QT prolongation.**

Benefits have been reported with **valproic acid (750–3000 mg/d), anticholinergics, or botulinum toxin injections.**

Other approaches include **baclofen (40–80 mg/d) or clonazepam (1–8 mg/d).**

In some cases, the abnormal movement is **refractory** to therapy.

Tardive Dystonia

Chronic neuroleptic exposure can also be associated with tardive dystonia, with **preferential involvement of axial muscles and characteristic rocking movements** of the trunk and pelvis.

Tardive dystonia is more troublesome than tardive dyskinesia and **frequently persists** despite stopping medication.

Valproic acid, anticholinergics, and botulinum toxin may occasionally be beneficial, but patients are **frequently refractory** to medical therapy.

Neuroleptic Malignant Syndrome

Neuroleptic medications can be associated with a neuroleptic malignant syndrome (NMS).

NMS is characterized by the **acute or subacute onset** of muscle rigidity, elevated temperature, altered mental status, hyperthermia, tachycardia, labile blood pressure, renal failure, and markedly elevated creatine kinase levels.

Symptoms typically evolve **within days or weeks** after initiating the drug. NMS can also be precipitated by the abrupt withdrawal of dopaminergic medications in PD patients.

Neuroleptic Malignant Syndrome

Treatment involves **immediate cessation** of the offending antipsychotic drug and the introduction of a **dopaminergic agent (e.g., a dopamine agonist or levodopa), dantrolene, or a benzodiazepine.**

When oral intake is not possible, **rotigotine subcutaneously or an infusion pump (delivering apomorphine subcutaneously)** may be used.

Treatment includes supportive measures such as **control of body temperature**, **hydration, electrolyte replacement, and control of renal function and blood pressure.**

Serotonin syndrome

Drugs that have serotonin-like activity (tryptophan, MDMA or “ecstasy,” meperidine) or that **block serotonin reuptake** can induce a rare, but potentially fatal, serotonin syndrome that is characterized by confusion, hyperthermia, tachycardia, and coma as well as rigidity, ataxia, and tremor.

Myoclonus is often a prominent feature, in contrast to NMS, which it resembles in other respects.

Patients can be managed with **propranolol, diazepam, diphenhydramine, chlorpromazine, or cyproheptadine as well as supportive measures.**

Drug-induced Movement Disorders

Phenytoin	**Chorea, Dystonia, Tremor, Myoclonus**
Carbamazepine	**Tics, Dystonia**
Tricyclic Antidepressants	**Dyskinesias, Tremor, Myoclonus**
Fluoxetine	**Myoclonus, Chorea, Dystonia**
Oral Contraceptives	**Dyskinesias**
Beta Adrenergics	**Tremor**
Buspirone	**Akathisia, Dyskinesia, Myoclonus**
Digoxin, Cimetidine, Diazoxide, Lithium, Methadone, And Fentanyl	**Dyskinesias**

Paroxysmal Dyskinesias

Rare disorders characterized by **episodic, brief involuntary movements** that can manifest as various types of hyperkinetic movements, including chorea, dystonia, tremor, myoclonus, and ballism.

There are three main types:

1. **Paroxysmal kinesigenic dyskinesia (PKD),** where the involuntary movements are triggered by sudden movement.
2. **Paroxysmal nonkinesigenic dyskinesias (PNKD),** where the attacks are not induced by movement.
3. **Exertion-induced dyskinesia (PED),** where attacks are induced by prolonged exercise.

Restless Legs Syndrome (Rls)

RLS is a neurologic disorder that affects ~10% of the adult population (it is **rare in Asians**). **Only, recently recognized as being a movement disorder.**

The **four core symptoms** required for diagnosis are as follows: **an urge to move the legs usually caused or accompanied by an unpleasant sensation in the legs; symptoms that begin or worsen with rest; partial or complete relief by movement; and worsening during the evening or night.**

Restless Legs Syndrome (Rls)

Symptoms **most commonly** begin in the legs, but can spread.

In about 80% of patients, RLS is associated with **periodic leg movements (PLMs)** during sleep and occasionally while awake.

These involuntary movements are **usually brief, lasting no more than a few seconds, and recur every 5–90 s.**

Restless Legs Syndrome (Rls)

The restlessness and PLMs are a major cause of **sleep disturbance** in patients, leading to poor-quality sleep and daytime sleepiness.

Primary RLS has a strong genetic component, an **autosomal dominant** pattern of inheritance.

The mean age of onset in familial forms is 27 years, although pediatric cases are recognized.

The severity of symptoms is variable.

Restless Legs Syndrome (Rls)

Secondary RLS may be associated with pregnancy or a range of underlying disorders, including anemia, ferritin deficiency, renal failure, and peripheral neuropathy.

The pathogenesis probably involves **disordered dopamine function,** which may be peripheral or central, possibly in association with an **abnormality of iron metabolism.**

Diagnosis is made on clinical grounds but can be **supported by polysomnography and the demonstration of PLMs.**

The neurologic examination is **normal.** Secondary causes of RLS should be excluded, and ferritin levels, glucose, and renal function should be measured.

Restless Legs Syndrome (Rls)

Most RLS sufferers have mild symptoms that **do not** require specific treatment.

If symptoms remain intrusive, low doses of dopamine agonists, e.g., pramipexole (0.25–0.5 mg), ropinirole (1–2 mg), or patch rotigotine (2–3 mg), taken 1–2 h before bedtime.

Levodopa may also be effective but is associated with unpleasant side effects.

Other drugs that can be effective include **anticonvulsants, analgesics, and opiates.**

Management of secondary RLS should be directed to **correcting the underlying disorder**; for example, iron replacement for anemia.

Other Disorders That May Present With A Combination Of Parkinsonism And Hyperkinetic Movements

WILSON'S DISEASE

WD is an **autosomal recessive** inherited disorder of copper metabolism that manifests with **neurologic, psychiatric, and liver disorders,** alone or in combination.

Neurologic onset usually manifests with **tremor, rigidity, and dystonia.**

Other features can include **parkinsonism with bradykinesia, dystonia (particularly facial grimacing), dysarthria, and dysphagia.**

More than half of those with neurologic features have a history of **psychiatric disturbances,** including depression, mood swings, and overt psychosis.

WILSON'S DISEASE

Diagnosis

Presence of Kayser-Fleischer (deposition of copper in Descemet's membrane) rings.

Low levels of blood copper and ceruloplasmin and high levels of urinary copper.

MRI reveals **generalized brain atrophy** and ~50% have signal **hypointensity** in the **caudate head, putamen, globus pallidus, substantia nigra, and red nucleus** on T2-weighted MRI scans.

Liver biopsy with demonstration of high copper levels and genetic testing remain the gold standard for the diagnosis.

WILSON'S DISEASE

Treatment is directed at **reducing tissue copper levels** and maintenance therapy to **prevent reaccumulation.**

Penicillamine can be used to increase copper excretion, but may lead to a worsening of symptoms in the initial stages of therapy. Side effects are common and can be attenuated by coadministration of pyridoxine. **Tetrathiomolybdate** blocks the absorption of copper and can be used instead of penicillamine. **Trientine and zinc** are useful drugs for maintenance therapy.

Patients with advanced hepatic disease may require a **liver transplant.**

Neurodegeneration With Brain Iron Accumulation

NBIA represents a group of inherited disorders characterized by **iron accumulation in the basal ganglia.**

Clinically, they can manifest as a progressive neurologic disorder with a variety of clinical features including **parkinsonism, dystonia, neuropsychiatric abnormalities, and retinal degeneration.**

Cognitive disorders and cerebellar dysfunction also seen.

Presentation is **usually in childhood**, but adult cases have been described.

Neurodegeneration With Brain Iron Accumulation

Pantothenate kinase–associated neurodegeneration (PKAN) caused by a mutation in the PANK2 gene, and is the most common form of NBIA.

MRI shows a characteristic **low signal abnormality in the center of the globus pallidus** on T2-weighted scans caused by iron accumulation known as the “eye of the tiger” sign.

Not all cases with iron accumulation in the basal ganglia represent an NBIA, because **iron accumulation** in specific basal ganglia regions is **normal**.

Psychogenic (Functional) Disorders

All movement disorders can be psychogenic in origin.

Tremor affecting the **upper limbs is the most common** psychogenic movement disorder.

Psychogenic movements can result from a **somatoform or conversion disorder, malingering (e.g., seeking financial gain), or a factitious disorder (e.g., seeking psychological gain).**

Psychogenic movement disorders are **relatively common, more frequent in women**.

Psychogenic (Functional) Disorders

Clinical features suggesting a psychogenic movement disorder include an **acute onset** with a pattern of abnormal movement that is **inconsistent with a known movement disorder.**

Diagnosis is based on the **nonorganic quality of the movement**, the absence of findings of an organic disease process, and positive features that specifically point to a psychogenic illness such as **variability and distractibility.**

For example, the magnitude of a psychogenic tremor is increased with attention and diminishes or even disappears when the patient is distracted by being asked to perform a different task or is unaware that he or she is being observed.

Psychogenic (Functional) Disorders

Other **positive features** suggesting a psychogenic problem include a **tremor frequency that is variable** or that entrains with the frequency of a designated movement in the contralateral limb, or a response to placebo interventions.

Associated features can include nonanatomic sensory findings, give-way weakness, astasia-abasia (an odd, gyrating gait or posture) and multiple somatic complaints with no underlying pathology (somatoform disorder).

Comorbid psychiatric problems such as **anxiety, depression, and emotional trauma** may be present.

Psychogenic (Functional) Disorders

The diagnosis can often be made based on **clinical features alone**.

Psychotherapy and hypnosis may be of value for patients with conversion reaction, and **cognitive behavioral therapy** may be helpful for patients with somatoform disorders.

Patients with hypochondriasis, factitious disorders, and malingering have a poor prognosis.

TABLE 428-1 Hyperkinetic Movement Disorders	
Tremor	Rhythmic oscillation of a body part due to intermittent muscle contractions
Dystonia	Involuntary, patterned, sustained, or repeated muscle contractions often associated with twisting movements and abnormal posture
Athetosis	Slow, distal, writhing, involuntary movements with a propensity to affect the arms and hands (this represents a form of dystonia with increased mobility)
Chorea	Rapid, semi-purposeful, graceful, dance-like nonpatterned involuntary movements involving distal or proximal muscle groups. When the movements are of large amplitude and predominant proximal distribution, the term *ballism* is used.
Myoclonus	Sudden, brief (<100 ms), jerk-like, arrhythmic muscle twitches
Tic	Brief, repeated, stereotyped muscle contractions that can often be suppressed for a short time. These can be simple and involve a single muscle group or complex and affect a range of motor activities.

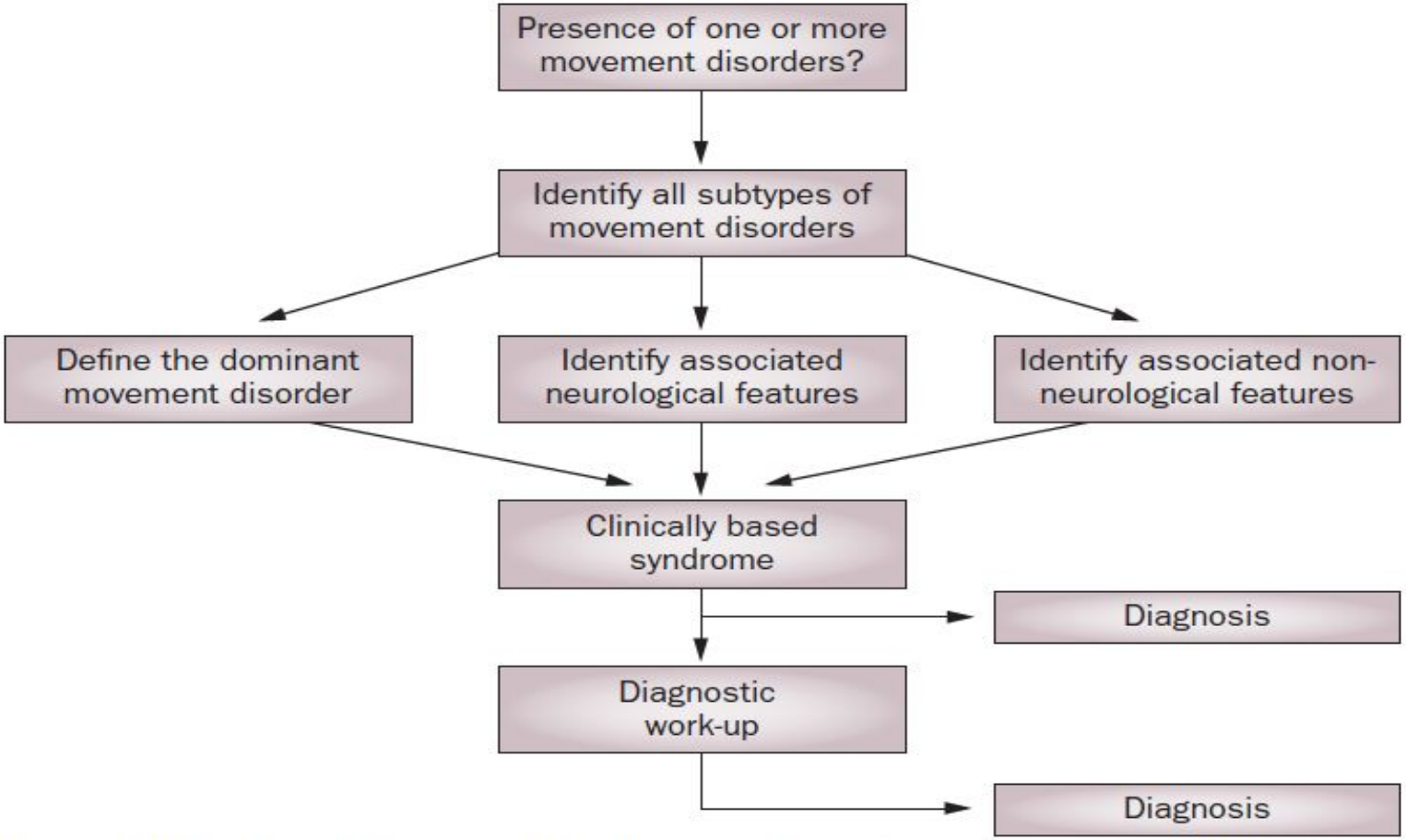


Figure 1 | A systematic approach to diagnosis in patients presenting with movement disorders.

Thank You